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EXAMINER	
SHALU, ZELALEM W	

ART UNIT	PAPER NUMBER
2145	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/558,191	Applicant(s) SAKURAI et al.	
	Examiner ZELALEM W SHALU	Art Unit 2145	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10/31/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 10/31/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date See Continuation Sheet.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Continuation of Attachment(s) 2) Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date: 10/31/2023,11/17/2023, 10/23/2024 AND 07/13/2026

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. This action is in response to the Application filed on 10/31/2023. Claims 1-19 are pending in the case.

Priority

3. The present application claims priority under 35 U.S.C. §119 to JP patent Application No. 2011-080622, filed on 2011-05-11. Applicant's claim for the benefit of a prior-filed application under 35 U.S.C. 119 and/or 35 U.S.C. 120 is acknowledged.

Information Disclosure Statement

4. As required by MPEP 609 (c), the Applicants' submission of the Information Disclosure Statement(s) filed on 10/31/2023, 11/17/2023, 10/23/2024 AND 07/13/2026 are acknowledged by the examiner and the cited references have been considered in the examination of the claims now pending.

Claim Interpretation

The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

5. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function.

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation(s) is/are:

Claim 1 recites:

... an estimation section that inputs a physical quantity ...; and ... an estimation section that estimates a physical quantity."

Claim 9 recites:

... function as an estimation section that ...."

Claim 10 recites:

... an acquisition section that acquires ...

... a learning model generation section that generates ... results of the acquisition by the acquisition section."

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

As per Applicant's specifications:

As for claim 1 and 9, specification provides sufficient structure and algorithm to perform the claimed function. Fig.1, [0053], discloses the estimation section 5 structure that include "the elastic body performance estimation device 1 includes an estimation section 5. First input data 3 indicating a magnitude of a pressure on the rubber actuator 2 (a pressure characteristic by operation cycling) are input as pressure data to the estimation section 5. Moreover, second input data 4 indicating a magnitude of an electrical characteristic of the rubber actuator 2 (the electrical characteristic by operation cycling) are input as electrical resistance data. The estimation section 5

outputs, as performance data, output data 6 indicating an estimation result of a performance state (performance indicator) of the rubber actuator 2". The estimation section 5 includes a trained learning model 51." In addition, Fig.14, [0076]-[0080], discloses a computer main body 100/CPU 102 programed to perform the estimation process using trained model 51.

As for claim 10, the specification provides sufficient structure and algorithm to perform the claimed function. Fig.8, [0049]- [0051], discloses the acquisition procedure "an example of the training data collection processing executed by the controller 70.".

If applicant does not intend to have this/these limitation(s) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Therefore, the dependent claims 2-7 and 11-19 are also interpreted under 35 U.S.C. 112(f) because they inherit the 112(f) interpretation of the "estimation section 'of claim 1."

Examiner Comments

8. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the

prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

Claim Rejections - 35 USC § 103

9. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

10. Claims 1-19 are rejected under 35 U.S.C. 103 as being unpatentable over Kim (Pub. No.: US 20070265808 A1, Pub. Date: 2007-11-15) in view of Pecht (Pub. No.: US 20100191681 A1, Pub. Date: 2010-07-29) in further view of Bryant (Pub. No.: US 20110046929 A1, Pub. Date: 2011-02-24)

Regarding independent Claim 1,

Kim teaches **an estimation device comprising an estimation section** (see Kim: fig.1, methods for monitoring structural health conditions**), that:**

- **inputs a physical quantity indicating a plurality of member characteristics for an estimation target member into a learning model** (see Kim: Fig.12, [0148], "1200 for identifying and determining the time-of-arrivals for Lamb wave modes in accordance with one embodiment of the present invention. In step 1202, the process module may load a set of sensor signal data from a signal database depository, such as computer 514, where each sensor signal data may be measured at one excitation frequency.")

- **a plurality of sets of a physical quantity indicating a performance state related to deformation of the member** (see **Kim**: Fig.14B, [0161], "a flow chart 1430 illustrating exemplary procedures for generating a tomographic image to identify regions having changes in structural conditions or damages in accordance with another embodiment of the present invention. In step 1432, the process module may load a time-of-arrival dataset of a Lamb wave mode, such as S.sub.o mode. As mentioned, the time-of-arrival for a Lamb wave mode can be used as a SCI. Using the extracted ridge curves in step 1212, the process module may exactly determine for all the network paths the time-of-arrival differences between the Lamb wave modes. Next, in step 1434, a conventional algebraic reconstruction technique may be applied to the loaded time-of-arrival dataset for the global inspection of damage on the host structure."), **and has been trained by being input with physical quantities indicating the plurality of member characteristics so as to output physical quantities indicating the performance state related to deformation of the member** (see **Kim**: Fig.15B, [0169], "the classification module may provide the SCI chromosome distribution 1524 on the grid points, best fitted to the artificial damage. With these SCI chromosomes, an unsupervised neural network can be trained in step 1526 to achieve the clustering or classification on the SCI distribution set on the grid points. However, the classification module can repeat to adapt the hybrid expert system while the process module process to renew the SCI distributions for each excitation frequency."), **and**
- **estimates a physical quantity indicating a performance state related to deformation of the estimation target member** (see **Kim**: Fig.18B, [0169], "the classification module may provide the SCI chromosome distribution 1524 on the grid points, best fitted to the artificial

damage. With these SCI chromosomes, an unsupervised neural network can be trained in step 1526 to achieve the clustering or classification on the SCI distribution set on the grid points. However, the classification module can repeat to adapt the hybrid expert system while the process module process to renew the SCI distributions for each excitation frequency.")

Pecht teaches the system wherein:

- **employs, as training data, a plurality of sets of a physical quantity indicating a plurality of member characteristics of different types which change in time series** (see **Pecht**: Fig.7, [0057], "Healthy data from historical healthy data or currently acquired data are chosen as training data 701. Special data from the training data 701 are picked to create memory matrix D (705). In one such approach, by way of example only, both the extreme data, that is the maximum and minimum values recorded within a data interval are selected, and combined with the recorded values at given occurrences, such as at every 5.sup.th position, where the values are arranged in ascending or descending order. The remaining training data L, that is the data not entered in memory matrix 705, is designated by box 703. When memory matrix D (705) is created, MSET goes through two processes. One, with reference to boxes 703 and 709, is to calculate estimates (L.sub.est) 709 of all of the remaining training data L 703 that were not chosen by the memory matrix 705 even though they are training data.")

Because both **Kim** and **Pecht** are in the same/similar field of endeavor of prognostics and health management (PHM) method for natural aging systems, accordingly, it would have been obvious to a person of ordinary skill in the art, before the effective filing date of the

invention, to modify the system, the teaching of **Kim** to include the system *that employs, as training data, a plurality of sets of a physical quantity indicating a plurality of member characteristics of different types which change in time series* as taught by **Pecht**. One would have been motivated to make such a combination to provide advanced warning of failures; can reduce the life cycle cost of a system by decreasing inspection costs, downtime, and inventory; and can assist in the design and logistical support of fielded and future systems (see **Pecht** [0008])

Kim and **Pecht** does not explicitly teach the system wherein a training data which change in time series according to deformation of a linearly- or non-linearly-deforming member.

However, **Bryant** teaches the system **wherein a training data which change in time series** (see **Bryant:** Fig.1, [0021], a nonlinear model 108 which processes the data from the input time series 102 and which produces an output time series 110 "The invention optionally includes a linear model 112 which also processes the data from the same input time series 102 and which produces an output time series 114."), **according to deformation of a linearly- or nonlinearly-deforming member** (see **Bryant:** Fig.1, [0021], "The invention includes a nonlinear model 108 which processes the data from the input time series 102 and which produces an output time series 110. The nonlinear model is assumed to be dependent on a number of adjustable parameters, making it able to approximate the effects of a wide range of structural characteristics including those with structural damage. The invention optionally includes a linear model 112 which also processes the data from the same input time series 102 and which produces an output time series 114.")

Because **Kim**, **Pecht** and **Bryant** are in the same/similar field of endeavor of damage detection or structural health monitoring, accordingly, it would have been obvious to a person of ordinary skill in the art, before the effective filing date of the invention, to modify the system, the teaching of **Kim** to include the system *that include a training data which change in time series according to deformation of a linearly- or nonlinearly-deforming member* as taught by **Bryant**. One would have been motivated to make such a combination to provide efficient and advanced warning of failures can reduce the life cycle cost of a system by decreasing inspection costs, downtime, and inventory improve the chances of successfully detecting distortion and identifying structural damage when it exists.

***Regarding Claim 2*,**

As shown above, **Kim**, **Pecht** and **Bryant** and teaches all the limitations of Claim 1. **Kim** further teaches the system wherein:

- **The member has an electrical characteristic that changes according to the deformation** (See **Kim**: Fig.1A, [0079], "Upon application of an electrical signal, the piezoelectric layer 116 may deform to generate Lamb waves. Also, the piezoelectric device 108 may operate as a receiver for sensing vibrational signals, converting the vibrational signals applied to the piezoelectric layer 116 into electric signals and transmitting the electric signals through the wires 118a-b. The wires 118a-b may be a thin ribbon type metallic wire.")
- **the physical quantity indicating the plurality of member characteristics includes a first physical quantity that indicates a pressure characteristic for deforming the member**

and a second physical quantity that indicates the electrical characteristic that changes according to the deformation of the member (see **Kim**: Fig.4A, [0111], "the diagnostic patch washer 400 may have a hollow space 403 to accommodate other fastening device, such as a bolt or rivet. FIG. 4C is a schematic diagram of an exemplary bolt-jointed structure 420 using the diagnostic patch washer 400 in accordance with one embodiment of the present invention. In the bolt-jointed structure 420, a conventional bolt 424, nut 426 and washer 428 may be used to hold a pair of structures 422a-b, such as plates. It is well known that structural stress may be concentrated near a bolt-jointed area 429 and prone to structural damages. The diagnostic patch washer 400 may be incorporated in the bolt-joint structure 420 and used to detect such damages.")

- **the physical quantity indicating the performance state related to deformation of the member includes a third physical quantity that indicates a number of repetitions of deformation of the member; and the learning model is trained so as to output the third physical quantity with the first physical quantity and the second physical quantity as input** (see **Kim**: Fig.4D, [0112], "To detect the structural damages near the bolt-joint area, a pair of diagnostic patch washers 400a-b may be inserted within the honeycomb portion 448, as illustrated in FIG. 4D. A sleeve 446 may be required to support the top and bottom patch washers 400a-b against the composite laminate layer 442. Also, a thermal-protection circular disk 444 may be inserted between the composite laminate layer 422 and the diagnostic patch washer 400b to protect the washer 400b from destructive heat transfer.")

***Regarding Claim 3*,**

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 2. **Kim** further teaches the system wherein:

- **the member includes an elastic body that is formed with a hollow inside and that generates a contraction force in a specific direction due to a pressurized fluid being supplied to the hollow inside** (see **Kim**: Fig.6AD, [0127], "a bridge box and transmission links are not shown in FIG. 6D. The patches 662 may be a device 502 or a sensor 522. In the system 660, the patches 662 and diagnostic patch washers 664 may detect the defects 672 by sending or receiving Lamb waves as indicated by arrows 670. Typically, the defects 672 may develop near the holes for the fasteners. The diagnostic patch washers 664 may communicate with other neighborhood diagnostic patches 662 that may be arranged in a strip network configuration, as shown in FIG. 6D. In one embodiment, the optical fiber coil sensors 330 and 340 may be used in place of the diagnostic patch washers 664.")
- **the first physical quantity is a pressure characteristic indicating a plurality of pressure values in a time series in a case in which the pressurized fluid is supplied and supply of the pressurized fluid is cancelled** (see **Kim**: Fig.2, [0096], "The vibrational displacement or strain of the host structure incurred by Lamb waves may be superimposed on the strain of the optical fiber cable 224. According to a birefringence equation, the reflection angle on the cladding surface of the optical fiber cable 224 may be a function of the strain incurred by the compression and/or tension.")

- **the second physical quantity is an electrical characteristic indicating a plurality of electrical resistance values in a time series of the elastic body, which change according to the first physical quantity** (see **Kim**: Fig.16A, [0107], "the shape and location of the hot-spot region 1610 may vary according to the excitation frequency and the number of network paths. Also, the diversity in physical characteristic and geometry of structures monitored may increase the difficulty level in classifying the damages. In one embodiment of the present invention, the classification module may employ a multilayer perception (MLP) or feedforward neural network to classify the damage of `hot-spot` region 1610 in a structure. T"); **and**
- **the third physical quantity is a performance indicator indicating a performance state for a plurality of respective repetition number groups that result from dividing a predetermined specific number of repetitions, as a physical quantity indicating a performance state of the member while deformable and sustaining a specific performance, into a plurality of levels** (see **Kim**: Fig.17AB, "illustrating exemplary procedures of a classification module to build a damage classifier using a codebook generated by the steps in FIG. 17A in accordance with one embodiment of the present invention. The damages may be located in a `hot-spot` region on the grid points of the diagnostic network paths. The SCI distribution 1734 of `hot-spot` regions for each structural condition may be used to design the codevector for structural conditions or damages, where each type of damage may belong to one of the types 1732. Each SCI distribution 1734 may be obtained at an actuation frequency."),

***Regarding Claim 4*,**

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 2. **Kim** further teaches the system wherein:

- **the member includes an elastic body that is formed with a hollow inside and that generates a contraction force in a specific direction due to a pressurized fluid being supplied to the hollow inside** (see **Kim**: Fig.6AD, [0127], "a bridge box and transmission links are not shown in FIG. 6D. The patches 662 may be a device 502 or a sensor 522. In the system 660, the patches 662 and diagnostic patch washers 664 may detect the defects 672 by sending or receiving Lamb waves as indicated by arrows 670. Typically, the defects 672 may develop near the holes for the fasteners. The diagnostic patch washers 664 may communicate with other neighborhood diagnostic patches 662 that may be arranged in a strip network configuration, as shown in FIG. 6D. In one embodiment, the optical fiber coil sensors 330 and 340 may be used in place of the diagnostic patch washers 664.");
- **the first physical quantity is a pressure characteristic indicating a plurality of pressure values in a time series in a case in which the pressurized fluid is supplied and supply of the pressurized fluid is cancelled** (see **Kim**: Fig.2, [0096], "The vibrational displacement or strain of the host structure incurred by Lamb waves may be superimposed on the strain of the optical fiber cable 224. According to a birefringence equation, the reflection angle on the cladding surface of the optical fiber cable 224 may be a function of the strain incurred by the compression and/or tension.")
- **the second physical quantity is an electrical characteristic indicating a plurality of electrical resistance values in a time series of the elastic body, which change according to**

the first physical quantity (see **Kim**: Fig.16A, [0107], "the shape and location of the hot-spot region 1610 may vary according to the excitation frequency and the number of network paths. Also, the diversity in physical characteristic and geometry of structures monitored may increase the difficulty level in classifying the damages. In one embodiment of the present invention, the classification module may employ a multilayer perception (MLP) or feedforward neural network to classify the damage of `hot-spot` region 1610 in a structure. T"); **and**

- **the third physical quantity is a product lifespan indicator that indicates a performance state from a performance state at the number of repetitions of deformation of the member until a predetermined specific number of repetitions as a performance state of the member while deformable and sustaining a specific performance** (see **Kim**: Fig.17AB, "Illustrating exemplary procedures of a classification module to build a damage classifier using a codebook generated by the steps in FIG. 17A in accordance with one embodiment of the present invention. The damages may be located in a `hot-spot` region on the grid points of the diagnostic network paths. The SCI distribution 1734 of `hot-spot` regions for each structural condition may be used to design the codevector for structural conditions or damages, where each type of damage may belong to one of the types 1732. Each SCI distribution 1734 may be obtained at an actuation frequency.")

***Regarding Claim 5*,**

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 1**. Kim** further teaches the system wherein:

- **wherein the learning model is a model generated by training using a recurrent neural network** (see **Kim**: Fig.18B, [0181], "schematically illustrates the architecture of a recurrent neural network 1830 for forecasting the future system matrix in accordance with one embodiment of the present invention")

Regarding Claim 6,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 1. **Kim** further teaches the system wherein:

- **wherein the learning model is a model generated by training using a network by reservoir computing** (see **Kim**: Fig.18B, [0181], "schematically illustrates the architecture of a recurrent neural network 1830 for forecasting the future system matrix in accordance with one embodiment of the present invention")

Regarding Claim 7,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 1. **Kim** further teaches the system wherein:

- **the learning model is a model generated by training using a network by physical reservoir computing employing a reservoir accumulated with a plurality of sets of a physical quantity indicating an operation state of the member, a plurality of sets of a physical quantity indicating a member characteristic that changes according to the deformation, and a plurality of sets of a physical quantity indicating a performance of the member** (see **Kim**: Fig.14B, [0161], "illustrating exemplary procedures for

generating a tomographic image to identify regions having changes in structural conditions or damages in accordance with another embodiment of the present invention. In step 1432, the process module may load a time-of-arrival dataset of a Lamb wave mode, such as S.sub.o mode. As mentioned, the time-of-arrival for a Lamb wave mode can be used as a SCI. Using the extracted ridge curves in step 1212, the process module may exactly determine for all the network paths the time-of-arrival differences between the Lamb wave modes.")

Regarding independent Claim 8, 9 and 10

Claim 8 is directed to a method claim, claim 9 is directed to non-transitory storage medium storing a program and claim 10 is directed to a device claim and the claims have similar/same claim limitation as Claim 1 and are rejected under the same rationale.

Regarding Claim 11,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 2. **Kim** further teaches the system wherein:

- **the learning model is a model generated by training using a recurrent neural network** (see **Kim**: Fig.18B, [0181], "schematically illustrates the architecture of a recurrent neural network 1830 for forecasting the future system matrix in accordance with one embodiment of the present invention")

Regarding Claim 12,

As shown above, **Kim** teaches all the limitations of Claim 3. **Kim** further teaches the system wherein:

- **the learning model is a model generated by training using a recurrent neural network** (see **Kim**: Fig.18B, [0181], "schematically illustrates the architecture of a recurrent neural network 1830 for forecasting the future system matrix in accordance with one embodiment of the present invention")

Regarding Claim 13,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 4. **Kim** further teaches the system wherein:

- **the learning model is a model generated by training using a recurrent neural network** (see **Kim**: Fig.16AC, [0179], "the prognosis module preferably utilizes a training method of recurrent neural network (RNN) with the previous dynamic reconstruction models determined from the simulated sensor signals, because of its highly nonlinear characteristics of the SCI vector I (. tau.). In an alternative embodiment,"

Regarding Claim 14,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 2. **Kim** further teaches the system wherein:

- **the learning model is a model generated by training using a network by reservoir computing** (see **Kim**: Fig.16AC, [0172], "To model the network dynamics of the diagnostic patch system, the prognosis module may compute the system matrix

E.sub..tau. (=[A.sub..tau.,B.sub..tau.,C.sub..tau.]) by using a subspace system identification method that reconstructs the dynamic system from the measured actuator/sensor signals in the network patches. The procedures disclosed by Kim et al., "Estimation of normal mode and other system parameters of composite laminated plates," Composite Structures, 2001 and by Kim et al., "Structural dynamic system reconstruction method for vibrating structures, Transaction of DSMC, ASME, 2003, which are incorporated herein in its entirety by reference thereto, can be employed to establish the reconstructed dynamic system model using the multiple inputs and outputs of the present sensory network system.")

Regarding Claim 15,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 3. **Thomas** further teaches the system wherein:

- **the learning model is a model generated by training using a network by reservoir computing** (see **Kim**: Fig.16AC, [0172], "a schematic diagram 1640 illustrating the fully connected network classifier for classifying a structural condition in accordance with one embodiment of the present invention. As illustrated in FIG. 16C, a set of Gabor jets 1642 may be generated using a SCI distribution 1643 that may contain 3 hot-spot regions 1641.")

Regarding Claim 16,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 4. **Kim** further teaches the system wherein:

- **the learning model is a model generated by training using a network by reservoir computing (**see **Kim**: Fig.16AC, [0172], "a schematic diagram 1640 illustrating the fully connected network classifier for classifying a structural condition in accordance with one embodiment of the present invention. As illustrated in FIG. 16C, a set of Gabor jets 1642 may be generated using a SCI distribution 1643 that may contain 3 hot-spot regions 1641.")

Regarding Claim 17,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 2. **Kim** further teaches the system wherein:

- **the learning model is a model generated by training using a network by physical reservoir computing employing a reservoir accumulated with a plurality of sets of a physical quantity indicating an operation state of the member, a plurality of sets of a physical quantity indicating a member characteristic that changes according to the deformation, and a plurality of sets of a physical quantity indicating a performance of the member** (see **Kim**: Fig.18A, [0177], "A structure suffers aging, damage, wear and degradation in terms of its operation/service capabilities and reliability. So, it needs a holistic view that the structural life has different stages starting with the elaboration of need right up to the phase-out. Given a network patch system, the current wave transmission of the network patch system may obey different time scales during the damage evolution to query the structure of its time-variant structural properties.")

Regarding Claim 18,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 3. **Kim** further teaches the system wherein:

- **the learning model is a model generated by training using a network by physical reservoir computing employing a reservoir accumulated with a plurality of sets of a physical quantity indicating an operation state of the member, a plurality of sets of a physical quantity indicating a member characteristic that changes according to the deformation, and a plurality of sets of a physical quantity indicating a performance of the member** (see **Kim**: Fig.18B, [0180], "estimate the future system matrix .SIGMA..sub..tau., the prognosis module preferably utilizes a training method of recurrent neural network (RNN) with the previous dynamic reconstruction models determined from the simulated sensor signals, because of its highly nonlinear characteristics of the SCI vector I(.tau.). In an alternative embodiment, the feed-forward neural network (FFN) can be used. The curves 1802 and 1810 may represent the evolution of the SCI vector I(.tau.) and the matrix .SIGMA..sub..tau., respectively, and span up to the time of structural death .tau..sub.e 1804. Sensor signals 1808 may be measured to access the structural conditions at time .tau..sub.v 1806.")

Regarding Claim 19,

As shown above, **Kim**, **Pecht** and **Bryant** teaches all the limitations of Claim 4. **Kim** further teaches the system wherein:

- **the learning model is a model generated by training using a network by physical reservoir computing employing a reservoir accumulated with a plurality of sets of a physical quantity indicating an operation state of the member, a plurality of sets of a physical quantity indicating a member characteristic that changes according to the deformation, and a plurality of sets of a physical quantity indicating a performance of the member** (see **Kim**: Fig.18B, [0181], "schematically illustrates the architecture of a recurrent neural network 1830 for forecasting the future system matrix in accordance with one embodiment of the present invention")

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

PGPUB NUMBER:	INVENTOR-INFORMATION:	TITLE / DESCRIPTION
US 20120133318 A1	Komatsu; Mayumi	**Title**: CONTROL APPARATUS, CONTROL METHOD, AND CONTROL PROGRAM FOR ELASTIC ACTUATOR DRIVE MECHANISM **Description**: The present invention relates to a control apparatus, a control method, and a control program for an elastic actuator drive mechanism, which serve for controlling operations of a drive mechanism that drives by an elastic actuator such as a fluid pressure drive actuator driven by deformation of an elastic body.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZELALEM W SHALU whose telephone number is (571)272-3003. The examiner can normally be reached M- F 0800am- 0500pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Cesar Paula can be reached at (571) 272-4128. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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